Using Machine Learning to Explore the Genetics of Tobacco Addiction: A Study of an African American Population

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Abstract

Cigarette smoking is a leading cause of preventable disease and death, responsible for over 7 million deaths annually. It contributes to various health problems, including cancer, cardiovascular diseases, respiratory infections, and diabetes. Understanding the genetic basis of tobacco addiction is key to developing effective therapies, with heritability estimates for smoking ranging from 21% to 84%. However, genetic variations alone explain only a small part of addiction, making it diffcult to predict smoking behavior. Machine learning (ML) can help overcome this by identifying complex associations between genetic markers and smoking status. In our study, we used ML tools like Elastic Net for feature selection and Logistic Regression for classification to explore the genetic factors of smoking in individuals of African origin, aiming to enhance prediction accuracy and uncover genes associated with nicotine dependence.

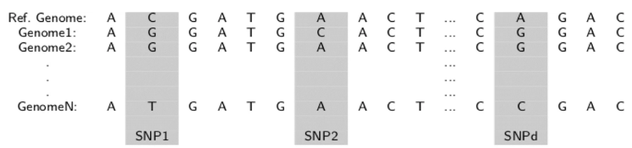
Introduction

The World Health Organization highlights that smoking-related lung cancer causes over 7 million deaths annually [1]. Smoking also increases the risk of various health problems, such as respiratory issues, ulcers, diabetes, and reproductive disorders. Many smokers want to quit, and understanding the genetic factors behind tobacco addiction is key to developing effective therapies. Smoking heritability ranges from 21% to 84%, suggesting that genetic factors can be identified [3].
This study explored machine learning (ML) techniques to classify smokers and non-smokers based on genetic data. Using a dataset from African-American participants, the data underwent cleaning, and feature selection was applied to identify relevant genetic variants. Methods such as support vector machines (SVM) and logistic regression were used to build classification models based on single nucleotide polymorphism (SNP) data. Tools like Elastic Net (EN) were used for feature selection, and functional annotation was performed to assess the relevance of identified genes to nicotine dependence.

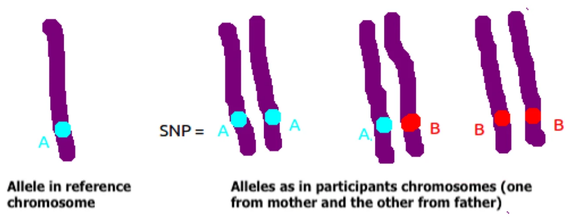
The results showed strong classification performance, particularly in logistic regression model, and identified SNPs could play a significant role in smoking prevention efforts.

Data

We retrieved data from the GEO platform under the identifier GSE148375 [5], which included 242,764 SNPs for 3,399 patients of African descent from the Mid-South Tobacco Case-Control (MSTCC) study.
The sample includes smokers, non-smokers and ex-smokers. These participants were divided into training (80%) and testing (20%) datasets.
After quality control, ambiguous SNPs were cleaned, and missing data was imputed using k-Nearest Neighbors (kNN). Genotypes were numerically encoded as 0, 1, or 2 based on minor allele counts, and SNPs were transformed into features for machine learning (ML) models. We further processed the data by creating binary variables for each SNP and split the dataset for model training and testing.



(a)



(b)

Figure 1. Illustration of three possible values of a SNP

Methods

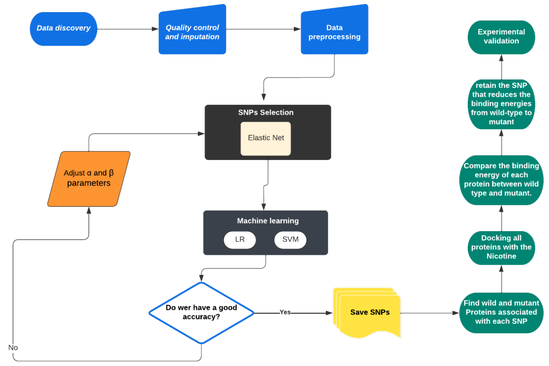


Figure 2. Illustration of the pipeline

Materials

We preprocessed the data to replace the SNPs by the values 0, 1 and 2 if there are 0, 1 or 2 mutations respectively. Then we reduced the SNPs with Elastic Net before using SVM and Logistic Regression for classification.

Elastic net:

$$\hat{\beta}_{\text{Ridge}} = \arg\min \left\{ \sum_{i=1}^{n} (y_i - \mathbf{x}_i\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}, \quad (1)$$

$$\hat{\beta}_{\text{Lasso}} = \arg\min \left\{ \sum_{i=1}^{n} (y_i - \mathbf{x}_i\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}, \quad (2)$$

$$\hat{\beta}_{\text{EN}} = \arg\min \left\{ \sum_{i=1}^{n} (y_i - \mathbf{x}_i\beta)^2 + \lambda((1-\alpha) \sum_{j=1}^{p} \beta_j^2 + \alpha \sum_{j=1}^{p} |\beta_j|) \right\}. \quad (3)$$

Logistic Regression (LR):

$$p(y=1|\mathbf{x}) = \frac{1}{1+e^{-\mathbf{w}^T\mathbf{x}}}, \quad (4)$$

$$\log\left(\frac{p(y=1|\mathbf{x})}{1-p(y=1|\mathbf{x})}\right) = \mathbf{w}^T\mathbf{x}. \quad (5)$$

Support Vector Machine (SVM):

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T\mathbf{x} + b), \quad (6)$$

$$\min_{\mathbf{w},b} \frac{1}{2}||\mathbf{w}||^2 \quad \text{subject to} \quad y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1, \quad \forall i. \quad (7)$$

Results

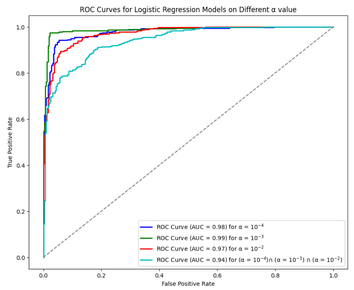


Figure 3. ROC of LR curves of different dataframes

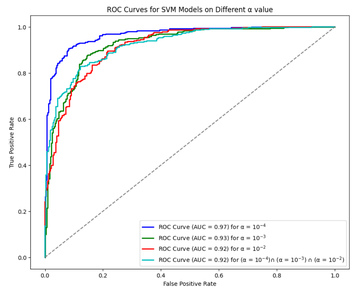


Figure 4. ROC of SVM curves of different dataframes

	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	94.11%	94.06%	94.18%	94.10%	98%
Support Vector Machine	91.02%	90.98%	91%	90.99%	97%

Table 1. For $\alpha = 10^{-4}$

	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	97.5%	97.46%	97.51%	97.49%	99%
Support Vector Machine	85.73%	85.70%	85.83%	85.71%	93%

Table 2. For $\alpha = 10^{-3}$

	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	91.02%	90.97%	91.09%	91.01%	97%
Support Vector Machine	82.79%	82.72%	82.73%	82.73%	92%

Table 3. For $\alpha = 10^{-2}$

	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	85.29%	85.34%	85.47%	85.28%	94%
Support Vector Machine	84.26%	84.41%	84.50%	84.26%	92%

Table 4. For $(\alpha = 10^{-4}) \cap (\alpha = 10^{-3}) \cap (\alpha = 10^{-2})$

Discussions

In our study, we identified 699 SNPs associated with 458 genes, including 14 olfactory receptor genes, aligning with prior research suggesting a link between olfactory receptors and smoking initiation [5]. While discussing each SNP individually is impractical, the identification of these olfactory receptor genes opens possibilities for future molecular docking studies to explore nicotine-receptor interactions, potentially leading to targeted therapeutic interventions. Additionally, we found the involvement of the CHRNA9 gene, known to play a role in nicotine dependence and smoking-related lung cancer [2], further supported by previous studies. Our findings also partially overlap with the Kyoto Encyclopedia of Genes and Genomes (KEGG), although integrating KEGG-identified SNPs into our model did not significantly improve classification performance, suggesting a level of concordance with our identified gene set.
Despite these promising results, there are limitations to consider. Our sample size was smaller than typical GWAS studies, though appropriate for our machine learning (ML)-focused approach. Additionally, the selection of a limited number of SNPs balanced practical constraints such as genotyping costs and computational resources. Our use of only two ML methods, SVM and Logistic Regression, also presents a limitation; exploring deep-learning approaches like Hyena-DNA [4] in future studies could yield valuable insights. However, the enrichment of nicotine-related biological processes through ClueGO and the functional annotation by DAVID validated our approach, supporting its potential for clinical applications.

Conclusion

This study utilized Support Vector Machine, Logistic Regression, and Elastic Net to develop a robust predictive model for smoking behavior, which may aid in smoking prevention efforts. Future research could explore the model's applicability across diverse ethnic populations to enhance its generalization.

References

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